

CE/CZ4055: CPS Security

Tutorial 1

(1) Charlie Chaplin Video: Is the eating machine a cyber physical system? If so, why is it not functioning properly? If not, why it is not a cyber physical system and what is missing to ensure that this is a cyber physical system.

(2)

(a) Construct a Mealy machine that calculates the 2's complement of a given input binary sequence. Assume that the bits are input to the state machine from the LSB to the MSB.

(b) Construct a Moore machine that calculates the number of occurrences of the word "abab" in any input string over a,b,c.

(c) For the Moore machine of 2(b), construct an equivalent Mealy machine.

(3) What are the Threats (T), Vulnerabilities (V) and Risks (R) for an autonomous vehicle?